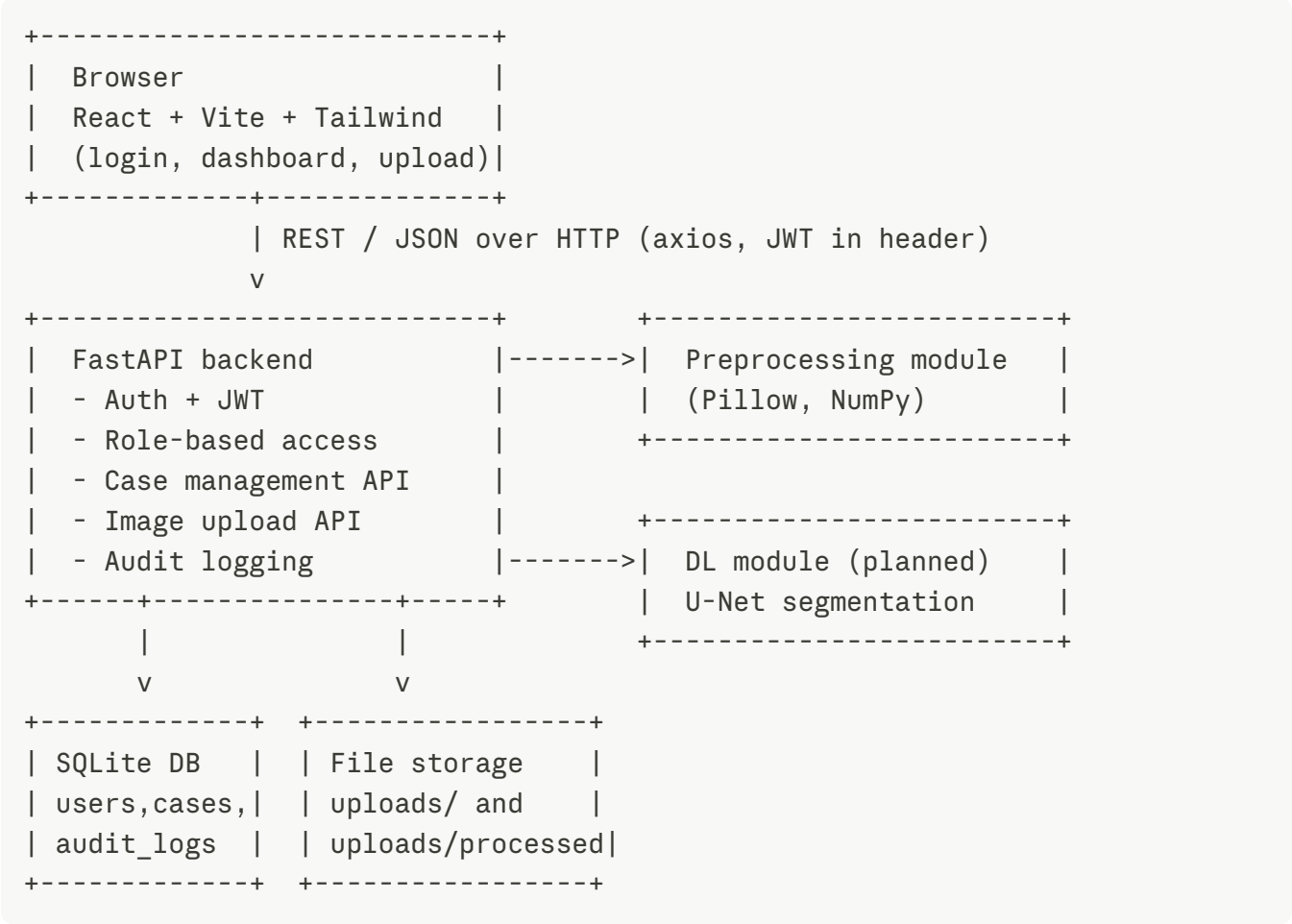
Artery Segmentation and Stenosis Detection Using Advanced DL Techniques

Artery AI | System Design Document: Architecture, Database Schema and UI Wireframes

1. System Architecture

Artery AI is a web application with a React frontend, a FastAPI backend, a SQLite database and local file storage for medical images. The current build delivers authentication, case management, image upload and a preprocessing pipeline. The deep learning models plug into the same backend.

1.1 Architecture diagram



1.2 Layers and technologies

Layer	Technology	Responsibility
Presentation	React, Vite, Tailwind CSS, axios	Login, register, dashboard, case forms, scan upload, preprocessing viewer
API	FastAPI, Uvicorn, CORS middleware	REST endpoints, validation, authentication
Security	JWT (python-jose), bcrypt	Token login, password hashing, role checks
Business logic	Python modules	Case management, preprocessing pipeline, audit logging
Data	SQLAlchemy, SQLite	Persistent storage of users, cases and audit logs
File storage	Local folder (uploads)	Original scans and processed pipeline images
AI (planned)	U-Net trained on the ARCADE dataset	Artery segmentation and stenosis detection in later stages

1.3 Main data flow

1. A user registers or logs in and receives a JWT token, which the frontend stores and sends with every request.
2. A clinician creates a case containing patient details and imaging information.
3. The clinician uploads an X-ray angiography image (PNG or JPG). The file is stored and linked to the case.
4. The preprocessing pipeline runs on request and returns six stage images for display.
5. Every create, update, delete, upload and preprocess action is written to the audit log.

1.4 Preprocessing pipeline

Raw image, then validation, normalization, noise reduction, contrast enhancement, region-of-interest extraction, and finally the 256 x 256 model input.

Stage	Method
Validation	File integrity check, grayscale conversion, minimum size check
Normalization	1st to 99th percentile intensity scaling to 0-255
Noise reduction	Median filter (3 x 3)
Contrast enhancement	Histogram equalization
ROI extraction	Crop away empty border around the imaged area
Model input	Resize to 256 x 256

1.5 Security design

- Passwords are hashed with bcrypt and never stored in plain text.
- Access tokens are JWTs that expire after 60 minutes.
- Role-based access control is enforced on the server for every case operation.
- Only de-identified or public datasets are used, with no real patient-identifiable data.
- An audit log records who did what and when.

2. Database Schema

The database has three tables. One user can own many cases, and one user can have many audit log entries.

```
users 1 ---- N cases
users 1 ---- N audit_logs
```

2.1 Table: users

Column	Type	Constraints
id	Integer	Primary key
name	String	Required
email	String	Required, unique, indexed

Column	Type	Constraints
hashed_password	String	Required
role	String	Default: clinician

2.2 Table: cases

Column	Type	Constraints
id	Integer	Primary key
owner_id	Integer	Foreign key to users.id, required
patient_id	String	Required
age	Integer	Optional
gender	String	Optional
medical_history	Text	Optional
risk_factors	Text	Optional
clinical_notes	Text	Optional
imaging_modality	String	Default: Coronary angiography
scan_date	String	Optional
anatomical_region	String	Optional
image_path	String	Set after upload
analysis_status	String	Default: Pending
model_version	String	Set after analysis (Milestone 2)
created_at	DateTime	Set automatically (UTC)

2.3 Table: audit_logs

Column	Type	Constraints
id	Integer	Primary key

Column	Type	Constraints
user_id	Integer	Foreign key to users.id, optional
user_name	String	Name of the acting user
role	String	Role of the acting user
action	String	Required, for example create_case, upload_scan
detail	String	Short description of the action
created_at	DateTime	Set automatically (UTC)

2.4 Roles and permissions

Role	Permissions
Radiologist / Clinician	Create, edit, delete and upload for their own cases; run preprocessing
Medical Researcher	View all cases, read-only
Data Scientist / ML Engineer	View all cases and run preprocessing, otherwise read-only
Administrator	Full access to all cases and the audit log

2.5 REST API summary

Method	Endpoint	Purpose
POST	/auth/register	Create an account
POST	/auth/login	Log in and receive a JWT
GET	/auth/me	Current user details
GET	/cases	List cases visible to the user
POST	/cases	Create a case
GET	/cases/{id}	View one case
PUT	/cases/{id}	Update a case

Method	Endpoint	Purpose
DELETE	/cases/{id}	Delete a case
POST	/cases/{id}/upload	Upload a scan image
POST	/cases/{id}/preprocess	Run the preprocessing pipeline
GET	/audit-logs	View the audit log (administrator only)

3. UI Wireframes

3.1 Login screen

```
+-----------------------------+-----------------------------+
|  Artery AI (dark panel)     |   Welcome back              |
|                             |   Email    [_____________] |
|  Intelligent coronary       |   Password [_____________] |
|  artery analysis            |            [Show]           |
|  ~~ heartbeat line ~~       |   [      Sign in         ]  |
|  - Artery segmentation      |   New here? Create account  |
|  - Stenosis detection       |                             |
|  - Explainable AI           |                             |
|  - Reports                  |                             |
+-----------------------------+-----------------------------+
```

3.2 Register screen

```
+-----------------------------+-----------------------------+
|  Artery AI (dark panel)     |   Create your account       |
|                             |   Full name [___________]   |
|  Role descriptions:         |   Email     [___________]   |
|  - Clinician                |   Role      [ v select    ] |
|  - Researcher               |   Password  [___________]   |
|  - Data scientist           |   strength  [====    ]      |
|  - Administrator            |   Confirm   [___________]   |
|                             |   [    Create account    ]  |
+-----------------------------+-----------------------------+
```

3.3 Dashboard (case management)

```
+----------+-------------------------------------------------+
| Artery AI| Case Management                    Name (role) U |
|          +-------------------------------------------------+
| [Cases]  | [Total cases] [Awaiting analysis] [With scan]    |
| Segment. |                                                  |
| Stenosis | [ Search...                       ] [+ New Case]  |
| Reports  | Patient ID | Age/Gender | Modality | Region |... |
|          | [img] P001 | 20 / F     | Coronary | LAD    |... |
| [Logout] |   Upload scan | Preprocess | Edit | Delete       |
+----------+-------------------------------------------------+
```

3.4 New case form (popup)

```
+-----------------------------------------+
| New Case                                |
| Patient ID [______]  Age [___]          |
| Gender [ v ]         Scan date [____]   |
| Imaging modality [_____________________]  |
| Anatomical region [____________________]  |
| Medical history  [____________________]  |
| Risk factors     [____________________]  |
| Clinical notes   [____________________]  |
|                    [Cancel] [Create case] |
+-----------------------------------------+
```

3.5 Preprocessing results (popup)

```
+-------------------------------------------------+
| Preprocessing pipeline                          |
| [original]   [normalized]   [denoised]          |
| [contrast]   [roi]          [model input]       |
|                                      [Close]    |
+-------------------------------------------------+
```

3.6 Planned screens

- Segmentation view: the X-ray with the predicted vessel mask overlaid.
- Stenosis view: highlighted narrowed regions with a severity estimate and an explainability heatmap.

- Reports and history: downloadable case reports and comparison of previous analyses.